Dylan Jiasen Liu

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Education

University of California, Irvine — Irvine, CA
B.S. Candidate in Computer Science and Applied Mathematics — *Expected Graduation 2028*
GPA: 4.0, Dean's Honors List

Experience

Full Stack Developer — Ongoing
ICS Student Council — *Irvine, CA*

- Made regular open source contributions to Ant Almanac for over 15,000 active users
- Worked 7+ hours a week resolving issues, completing code reviews, and attending meetings
- Openly communicated with team leads and fellow developers, providing time estimates and participating in project discussions

Research Intern — Summer 2026
University of California, Irvine — *Irvine, CA*

- Investigated methods for using large language models (LLMs) to generate and interact with SQL queries
- Conducted literature reviews and contributed to research evaluating LLM-based SQL generation
- Developed scoring criteria for evaluating the correctness and quality of LLM-generated SQL
- Worked with healthcare data models including OMOP and MIMIC

Learning Assistant, ICS 6B — Fall 2025
University of California, Irvine — *Irvine, CA*

- Assisted Boolean logic and discrete structures lectures for 300+ students
- Led 21 private review sessions covering Boolean logic and related discrete mathematics concepts

Projects

AirPiano | *Arduino UNO, Edge AI* — Winter 2026

- Built a physical self-playing MIDI keyboard using an Arduino UNO
- Wired and assembled hardware for automated key presses and optimized AI models for edge devices

SpeedStreets | *React, Vite, Express, Socket.io, Overpass QL* — Winter 2025

- Managed the team's GitHub repository and reviewed pull requests throughout development
- Implemented socket connections with Socket.io and Express and contributed to the React frontend
- Developed RESTful API requests to OpenStreetMap data using Overpass QL

React Arena | *React, Vite, Express, Socket.io* — Winter 2025

- Built a real-time multiplayer web game using React and Vite
- implemented client-server communication and live game-state synchronization using Express and Socket.io

Chopsticks Solver | *React, Minimax, Alpha-Beta Pruning, Memoization* — Fall 2024

- Developed an interactive React-based Chopsticks game with an algorithm for selecting strategic moves
- Implemented minimax search, alpha-beta pruning, and memoization to efficiently explore game states

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks/Libraries: React, Node.js, Express, Socket.io, TensorFlow, Vite, Next.js
Tools/Technologies: Git, REST APIs, Arduino